Development of Major Ethical Theories

Utilitarianism, Duty Ethics, Rights Ethics, and Virtue Ethics

Introduction

Ethical theories are methods used to decide whether an action is right or wrong. In engineering, these theories are important because engineers often face complex cases where law, company policy, profit, safety, and public welfare conflict.

According to Fleddermann, ethical theories help engineers understand ethical problems and make better professional decisions. Engineering ethics does not always give one perfect answer, so theories help us compare different choices and defend the final decision.

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1. Why Ethical Theories Developed

Ethical theories developed because human beings needed a clear way to judge right and wrong. People cannot depend only on feelings, personal opinion, or culture because different people may see the same situation differently.

Fleddermann explains that Western ethical thought came from ancient Greek philosophy and later developed through many centuries. He also says that non-Western cultures independently developed similar ethical principles. This shows that ethics is a universal human need, not only one culture's idea.

In engineering, ethical theories are needed because engineers make decisions that affect public safety, health, environment, and society. These theories help engineers go beyond minimum law and think about moral responsibility.

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2. How Ethical Theories Help Engineers

Ethical theories help engineers analyze a case from different angles.

They help answer questions such as:

- Which action gives the most benefit?
- Which duty must be followed?
- Whose rights are being affected?
- What would a good and honest engineer do?

Fleddermann explains that when different theories give different answers, each answer must be carefully examined and balanced. Generally, rights and duties should be given strong weight because individual rights should not be sacrificed only for social benefit.

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3. Utilitarianism

Utilitarianism judges an action by its consequences. An action is ethical if it produces the greatest overall good for the greatest number of people.

In engineering, this theory asks engineers to compare benefits and harms. If a design improves public welfare, reduces risk, and benefits many people, utilitarianism may support it.

Fleddermann connects this idea with cost-benefit analysis, where costs and benefits are compared. However, he warns that cost-benefit analysis is not a complete ethical tool because it may ignore subjective values like human life, natural beauty, fairness, and individual suffering.

Types of Utilitarianism

Act utilitarianism looks at one particular action and judges whether that action gives the best result.

Rule utilitarianism looks at general rules. It asks whether following a rule will usually produce the best result for society.

Example:
If a company wants to skip safety testing to save money, utilitarianism asks whether saving money is greater than the possible harm to users. Since public harm may be serious, utilitarianism would reject skipping safety tests.

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4. Deontology / Duty Ethics

Deontology means duty-based ethics. In Fleddermann's book, this is explained as **duty ethics**.

Duty ethics says an action is ethical if it follows moral duties, even if breaking the duty may bring some benefit. Important duties include being honest, being fair, keeping promises, and not causing harm.

Fleddermann explains that Immanuel Kant was a major supporter of duty ethics. Kant believed moral duties are fundamental, and ethical actions are those that can be written as universal duties such as "be honest" and "do not cause suffering."

Example:
An engineer must not falsify test reports. Even if false reports save the company money, lying is still unethical because honesty is a duty.

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5. Rights Ethics

Rights ethics says people have basic rights that others must respect. These rights may include the right to life, safety, liberty, property, privacy, and fair treatment.

Fleddermann explains that rights ethics was largely developed by John Locke. It says people have fundamental rights, and other people have a duty to respect them.

Rights ethics and duty ethics are closely connected. If one person has a right to safety, then engineers have a duty not to create unsafe products.

Example:
If a factory pollutes drinking water, it violates people’s right to health and safety. Even if the factory creates jobs, people’s rights cannot be ignored.

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6. Virtue Ethics

Virtue ethics focuses on the character of the person instead of only rules or results. It asks:
What would a good engineer do?

A virtue is a good character quality. Important engineering virtues include honesty, fairness, responsibility, courage, care, trustworthiness, and competence.

Fleddermann explains that virtue ethics is useful in research and engineering because honesty creates trust, while dishonesty creates doubt. False experimental results and failure to give proper credit are unethical because they show dishonest character.

Example:
If an engineer finds a dangerous fault, virtue ethics says the engineer should act with honesty and courage by reporting it, not hiding it.

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7. Comparison of Theories

Utilitarianism

Focuses on **results**.

Question:
Which action gives the greatest overall benefit?

Duty Ethics

Focuses on **duties**.

Question:
What duty must I follow?

Rights Ethics

Focuses on **rights of individuals**.

Question:
Whose rights are being protected or violated?

Virtue Ethics

Focuses on **character**.

Question:
What would a good, honest, responsible engineer do?

Conclusion

Major ethical theories developed to help people judge right and wrong in a clear and reasoned way. In engineering, they are especially important because engineers make decisions that affect public safety, environment, business, and society.

Utilitarianism focuses on consequences and overall benefit. Duty ethics focuses on moral duties such as honesty and fairness. Rights ethics focuses on respecting basic human rights. Virtue ethics focuses on good character. A responsible engineer should use all these theories together and choose the action that best protects public safety, honesty, rights, and professional responsibility.

How to Apply in Case Studies

In a case study, first identify the main ethical conflict, such as safety vs. profit or honesty vs. pressure from management.

Then apply each theory:

Utilitarianism: Compare benefits and harms.
Duty ethics: Identify duties such as honesty, safety, and fairness.
Rights ethics: Identify whose rights are affected.
Virtue ethics: Ask what a good engineer would do.

Finally, give a balanced decision. If theories conflict, give more importance to public safety, individual rights, and professional duties.

Short Case Study: Unsafe Product

Suppose an engineer discovers that a product may harm users, but the company wants to release it quickly to earn profit.

Utilitarianism: Releasing it may benefit the company, but possible harm to many users is greater, so it should not be released.
Duty ethics: The engineer has a duty to be honest and protect public safety.
Rights ethics: Users have a right to safety and truthful information.
Virtue ethics: A good engineer would act honestly and courageously.

Therefore, the engineer should report the risk, ask for proper testing, and refuse to approve the unsafe product.

Memorization Checklist

Remember ethical theories with:

R-D-R-C

R — Results: Utilitarianism judges consequences.
D — Duties: Duty ethics follows moral duties.
R — Rights: Rights ethics protects individual rights.
C — Character: Virtue ethics focuses on good character.